Journal of Hazardous Materials 507 (2026) 141820

Contents lists available at ScienceDirect

Journal of Hazardous Materials

journal homepage: www.elsevier.com/locate/jhazmat

Structuring large language models for chemical health risk reasoning in environmental and occupational exposure

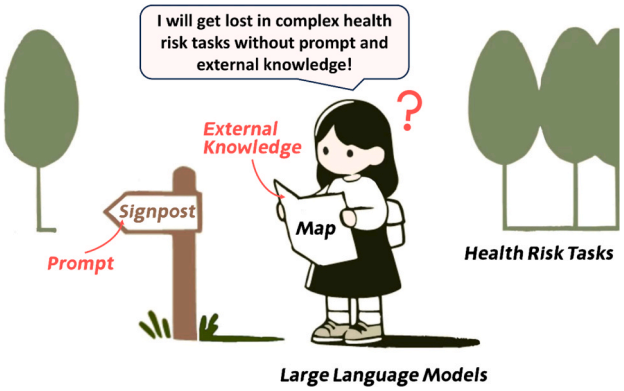
Zhuo Chen, Meng Du, Chen Qian *, Han-Qing Yu *

State Key Laboratory of Advanced Environmental Technology, Department of Environmental Science and Engineering, University of Science and Technology of China, Hefei 230026, China

HIGHLIGHTS

- A structured health risk assistant enables chemical reasoning through LLMs.
- RAG links compound properties with exposure and disease outcomes.
- Semantic structuring transforms fragmented data into coherent risk profiles.
- Domain-aware prompting reduces hallucination and improves interpretability.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Large language models
Semantic structuring
Domain-aware prompting
Occupational and environmental exposure
Chemical-induced disease risk

ABSTRACT

Large language models (LLMs) have revolutionized numerous scientific fields. However, their application in environmental health remains limited, as they are prone to hallucinations and struggle with reasoning over complex, real-world exposure data. Chemical risks, especially from rare or poorly studied compounds, span diverse occupational and everyday scenarios, where structured knowledge is often incomplete and poorly integrated. To address these challenges, we present a health risk assistant that integrates deep learning-based compound property prediction, retrieval-augmented generation (RAG), and prompt engineering into a unified reasoning framework. Central to our design are two synergistic strategies: semantic structuring, which reorganizes heterogeneous knowledge into context-consistent and semantically rich profiles, and domain-aware prompting, which guides large language models through structured and role-specific reasoning aligned with toxicological goals. The framework enables the system to map physicochemical attributes to exposure pathways and disease outcomes with improved relevance and fidelity. Evaluated on a 100-question benchmark spanning chronic toxicity, occupational hazards, and exposure-specific risks, our LLM-RAG system consistently outperforms baseline models in correctness, faithfulness, and domain-specific helpfulness. Beyond answering

* Corresponding authors.
E-mail addresses: qianc@ustc.edu.cn (C. Qian), hqyu@ustc.edu.cn (H.-Q. Yu).

https://doi.org/10.1016/j.jhazmat.2026.141820
Received 12 October 2025; Received in revised form 17 March 2026; Accepted 18 March 2026
Available online 19 March 2026
0304-3894/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

queries, our approach demonstrates how integrated knowledge curation and tailored prompting can enable trustworthy and scalable AI reasoning for public and environmental health protection.

1. Introduction

The rapid expansion of the chemical industry has introduced a growing variety of compounds into occupational and everyday environments. This has significantly increased the complexity of human exposure and raised new challenges in health risk assessment. These challenges arise from two main factors. One is the large number of chemical substances with limited toxicity data. Another is the complexity of environmental processes such as transport, transformation, and accumulation. These factors jointly affect pollutant behavior under real-world conditions. As a result, health risk assessment becomes particularly challenging. Conventional screening methods can assist in identifying unknown contaminants. However, they are often inadequate for assessing composite risks involving multiple compounds and evolving exposure scenarios.

To support more advanced understanding, artificial intelligence (AI) has become central to scientific research through the AI for Science (AI4S) paradigm [1]. In particular, machine learning (ML) techniques have been used to predict environmental behaviors and health implications of chemical exposures. Using experimental or theoretical datasets, researchers develop ML-based predictive models for compound properties. These models support ecotoxicological and environmental endpoint prediction. Traditional models, such as quantitative structure-activity relationships (QSAR) [2,3], along with modern deep learning (DL) models [4–6] such as graph neural networks (GNNs) [7,8] and transformers [9], have demonstrated efficacy in predicting toxicity endpoints [10], physicochemical properties, and environmental fate parameters [11] across a broad range of human and ecotoxicological applications [12]. Beyond single-task predictions, AI-based analyses are beginning to offer integrative insights that can support early disease prevention and inform exposure-related interventions [13].

More recently, large language models (LLMs) have transformed natural language processing by enabling both linguistic fluency and multi-domain reasoning. Highly parameterized models such as GPT-4 (OpenAI), LLaMA 3 (Meta), and DeepSeek-R1 are now capable of diverse tasks including generation, translation, and technical explanation. In healthcare [14–18] and chemical automation [19–22], LLMs assist with clinical decision support, molecular design, and automated synthesis planning. In environmental science, domain-aligned LLMs like TianGong GPT [23] and specialized environmental QA system [24], demonstrate promising capabilities tailored to environmental applications [25].

Despite recent advancements, applying LLMs to environmental health remains a significant challenge. Real-world exposure scenarios span both occupational and everyday environments. They involve diverse chemical classes, exposure pathways, and health outcomes. The complexity makes it difficult to establish reliable links between compound properties and disease risks. LLMs possess strong capabilities for integrating heterogeneous information and modeling complex associations. They are well-suited to map interactions among chemical properties, exposure activities, and health effects. However, a persistent limitation is hallucination, where generated outputs may deviate from factual knowledge or present misleading reasoning [26–28]. The issue is particularly concerning when dealing with rare compounds or poorly characterized toxicological outcomes, which are common in occupational health and environmental exposure contexts. In safety-relevant contexts, models can hallucinate clinically or environmentally relevant facts, misinterpret quantitative information, or provide overconfident but incorrect recommendations. Balancing factual accuracy with generative flexibility remains difficult, though emerging approaches such as data augmentation and prompt engineering offer potential. In addition, interpreting environmental health data requires domain-specific knowledge, which is often inaccessible to general-purpose models or non-expert users. While injecting curated knowledge into model inputs can improve relevance, fine-tuning remains resource-intensive and lacks scalability.

To overcome these limitations, we developed a domain-adaptive retrieval-augmented generation (RAG) framework for health risk reasoning that combines semantic structuring, MolCLR-based physicochemical property prediction, and domain-aware prompting. The system integrates a LLM with an external knowledge base encompassing physicochemical properties, environmental behavior, and disease associations. The structure enables the model to retrieve relevant evidence and produce scenario-specific assessments while minimizing hallucinations. Built on this foundation, we implemented a locally deployed LLM-RAG assistant with four core modules: knowledge processing, retrieval, generation, and evaluation (Fig. 1). To enrich the knowledge base, deep learning models predict key physicochemical parameters for poorly characterized compounds. These predictions enhance the system's ability to analyze both chronic and acute exposures across occupational and non-occupational scenarios. To improve the quality and relevance of outputs, we introduced two synergistic strategies. Semantic structuring reorganizes heterogeneous information into coherent, compound-level profiles to support context-consistent retrieval. Domain-aware prompting guides the model through structured reasoning using expert-role instructions and exposure-specific templates. For evaluation, we combined the LLM-as-a-judge paradigm with the RAGAS framework. The dual approach assesses correctness, helpfulness, informativeness, relevancy, and faithfulness using both model- and retrieval-based metrics. The system enables transparent and scalable LLM reasoning across diverse chemical health scenarios and offers a pathway for developing advanced tools for occupational disease prevention and environmental health protection.

2. Materials and methods

2.1. Local knowledge base construction

To support reliable health risk reasoning with LLMs, we constructed a domain-specific knowledge base using a semantic structuring strategy. The approach combines data augmentation, attribute alignment, and knowledge reorganization to enhance the consistency and informativeness of retrieved content.

We first extracted relevant information from environmental and healthcare-related web sources, focusing on human activities, compounds, environmental effects, and disease risks. The process yielded 12,203 unique knowledge entities, which were compiled into a semi-structured, expandable, and updatable knowledge base. These entities were extracted from a curated occupational toxicology and exposure database. The database aggregates authoritative scientific and regulatory sources, including U.S. occupational and environmental risk resources as well as international chemical safety portals. In our work, we used a static snapshot, and future deployments can refresh the vector store periodically as new data become available. Data engineering techniques were applied throughout to increase information density and improve semantic coherence.

Textual sources alone often lack the physicochemical specificity needed for compound-level reasoning. To address this limitation, we developed a structured data augmentation pipeline that reformats domain-specific datasets into chemically enriched profiles. Through field extraction and textual restructuring, the pipeline aligns both measured and predicted heterogeneous attributes into unified records

designed for LLM interpretability. These records enable the model to synthesize and reason across qualitative and quantitative dimensions.

To supplement missing data for rare or under-characterized compounds, we trained a deep learning model to predict key physicochemical properties. Specifically, MolCLR [29], an unsupervised pre-trained graph neural network trained on a curated PubChem subset [30], was fine-tuned using environmental and health risk datasets. The resulting model generated domain-adapted property predictions for incorporation into the knowledge base.

Eight physicochemical parameters were selected based on toxicological relevance and data availability: hydrogen-bond acceptor (HBA) count, topological polar surface area (TPSA), vapor pressure (PLV), aqueous solubility (SW), organic carbon–water partition coefficient (KOC), Henry's law constant (KH), lipophilicity (Lipo), and mutagenicity index (Muta). These indicators characterize compound transport, persistence, bioaccumulation, and potential health risks. Full rationale and parameter definitions are provided in Text S1.

2.2. LLM answer generation

To reduce hallucinations and improve response quality in health risk reasoning, we implemented two core strategies during generation: domain-aware prompting and inference parameter optimization. We designed a modular prompt template tailored for health risk tasks. The template specifies the model's expert role, incorporates scenario-relevant context, poses focused questions, and constrains response style. These elements jointly support interpretability, factual grounding, and alignment with user expectations. Domain-aware prompting further embeds hidden CoT cues to support step-wise reasoning, enhancing response depth and consistency. Small prompt adjustments yielded measurable improvements in brevity, relevance, and numerical precision. The pipeline is deployed locally using DeepSeek-R1:32B via Ollama, ensuring secure and cost-effective operation.

We fine-tuned inference parameters to balance creativity and accuracy. The temperature was set to 1, a commonly used setting in language generation tasks. The choice preserves output diversity while maintaining coherence, which is important for health risk reasoning in environmental exposure. It allows for the generation of diverse possibilities without deviating from factual knowledge. Additionally, top_p was set to 1. This setting avoids truncating the candidate token space and helps preserve specialized and rare terms when needed.

2.3. Evaluation of responses

Evaluating LLM-based chatbots is particularly challenging in specialized domains. Standard benchmarks often fail to capture task complexity or user expectations. Conventional metrics like BLEU, ROUGE, and BERTScore are less effective in open-domain, multi-turn dialogue due to their limited sensitivity to semantic fidelity and reasoning quality.

To address this, we employed a dual evaluation strategy. First, we applied the open-source RAGAS framework using GPT-4o-mini to quantify faithfulness and answer relevancy based on alignment between answers and retrieved knowledge. Second, we adopted the LLM-as-a-judge framework to assess correctness, usefulness, and informativeness, using GPT-o3 as an independent evaluator. AI judge outputs have been shown to align with human preference over 80% of the time [31], thereby reducing reliance on costly manual annotation. The judge model was instructed to ignore response length, position, and assistant identity to mitigate bias. Evaluation criteria and prompt are detailed in Text S2 and Fig. S1.

To assess the effect of individual components, we designed control-variable experiments on both data and prompt engineering. Six assistants were compared: a baseline LLM, the final LLM-RAG, and four LLM-RAG variants with ablated modules, including models without role specification ("Without Role"), with simplified context prompting ("Simplified Context"), without format guidance ("Without Format"), and using a knowledge base without semantic structuring ("Without

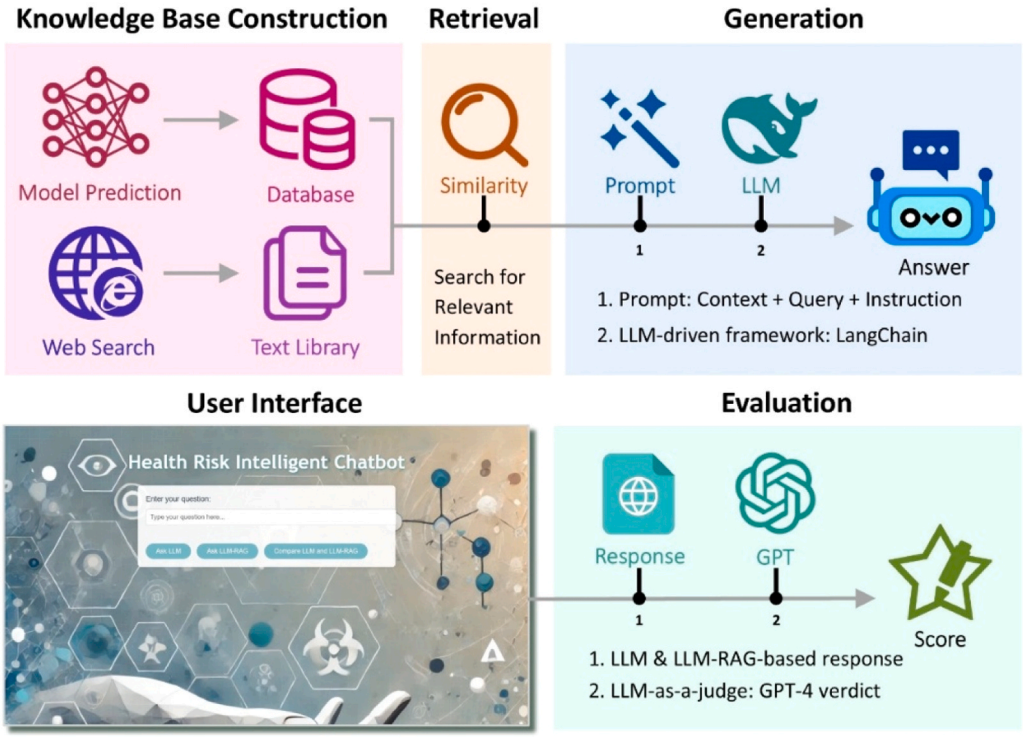


Fig. 1. Integrated framework of the health risk reasoning system based on LLM-RAG.
The workflow includes four key components: knowledge base construction via semantic structuring, retrieval-enhanced response generation guided by domain-aware prompting, interactive user interface, and performance evaluation through a dual assessment strategy combining RAGAS metrics and LLM-as-a-judge scoring.

Restructuring").

We developed a 40-question benchmark spanning nine environmental health topics: environmental fate and pollution pathways, physicochemical property-driven health risks, scenario-specific exposure controls, chronic exposure and cumulative toxicity, occupational chemical hazards, population-level environmental health risks, pharmacokinetic and toxicokinetic profiles, molecular structure and reactivity risks, and radiological and medical substance risks. These 40 questions were used for the retrieval-grounded evaluation with RAGAS. For the subsequent LLM-as-a-judge evaluation of reasoning quality, we expanded the benchmark to 100 questions by adding a 60-question broad-coverage set sampled from knowledge base entities to increase chemical-class breadth. Together, these topics were chosen to collectively cover the four canonical components of chemical risk assessment, including hazard identification, exposure assessment, dose-response assessment, and risk characterization. The benchmark also covers data-scarce situations, as well as multi-chemical combined exposure scenarios that reflect concurrent exposures typical in real occupational and environmental settings. Question types and examples are summarized in Text S3.

To strengthen scientific validity beyond automated judging, we conducted a human-in-the-loop expert validation. Three environmental health experts scored a subset of 30 benchmark questions using the same rubric as the AI-judge. We aggregated expert scores and quantified agreement with the LLM-as-a-judge scores using Pearson correlation.

2.4. LLM integration and deployment

We used LangChain to build and manage the retrieval-augmented generation pipeline. LangChain provides modular components for constructing LLM-based applications, facilitating smooth coordination of retrieval, generation, and evaluation tasks. It enabled efficient information extraction and management, supporting a flexible, extensible chatbot architecture. The full chain workflow is provided in Text S4. In our experiments, the system was deployed on a single on-premise workstation equipped with two NVIDIA RTX 3090 GPUs (2 × 24 GB VRAM), which was sufficient for interactive inference with the 32B model.

To ensure usability, we implemented a simple web-based chatbot interface that accepts natural-language questions and displays answers in a concise and structured format. The interface is intended for non-technical users such as enterprise safety managers and local

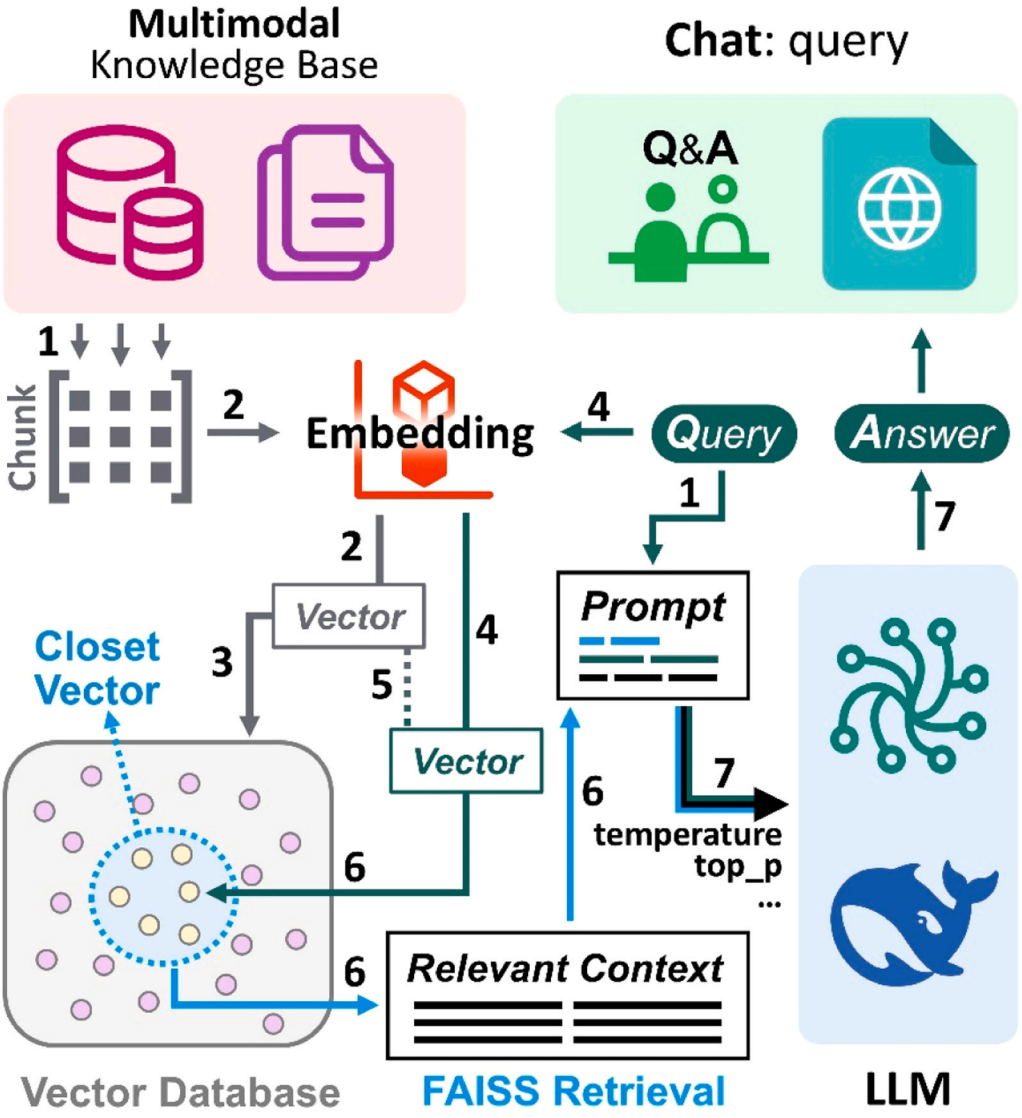


Fig. 2. Stepwise architecture of the chatbot system based on semantic structuring and prompt-guide retrieval. Steps 1–7 illustrate the full reasoning cycle: (1) Chunking, (2) Embedding, (3) Indexing, (4) Query Encoding, (5) Similarity Matching, (6) Prompt Construction, and (7) LLM Generation. The process enables context-grounded and semantically consistent answers to user queries.

environmental health personnel, providing actionable chemical safety information for both occupational and everyday exposure scenarios.

3. Results and discussion

3.1. Framework design for health risk reasoning

The health risk chatbot framework comprises three core components: a knowledge base, a retrieval mechanism, and a generation module (Fig. 2). The knowledge base integrates two types of information. The first is a structured database containing molecular properties predicted by a deep learning model that was fine-tuned after pretraining. The second is a semi-structured text repository that includes descriptions of hazardous compounds and their associated diseases as encountered in daily life. To make these sources usable for the language model, we applied data cleaning and synthesis procedures that reformat the content into a consistent and interpretable structure.

To facilitate efficient retrieval, the knowledge base is segmented into smaller text blocks (Step 1, Fig. 2). The chunking strategy reduces computational overhead and improves retrieval precision. Since raw text often includes ambiguity or noise, all content is converted into dense vector representations using an embedding model (Step 2, Fig. 2). These vectors preserve semantic relationships and contextual relevance. During system initialization, the RAG framework constructs a vector database from these embeddings (Step 3, Fig. 2). This index supports fast similarity-based retrieval in subsequent queries without re-embedding, allowing for efficient updates and dynamic scalability. The response generation process follows a three-stage sequence: retrieval, augmentation, and generation. Upon receiving a user query, the system encodes it into a vector using the same embedding model (Step 4, Fig. 2) and matches it against the vector index to identify the most relevant knowledge blocks (Step 5, Fig. 2). These blocks are then integrated into the LLM's context window (Step 6, Fig. 2), providing the background needed for informed reasoning. Finally, the retrieved content and user query are embedded into a structured prompt (Step 7, Fig. 2), guiding the LLM to produce targeted, context-aware responses. Prompt design is a critical component of the framework, as it directly shapes how the model interprets retrieved information and constructs reliable outputs.

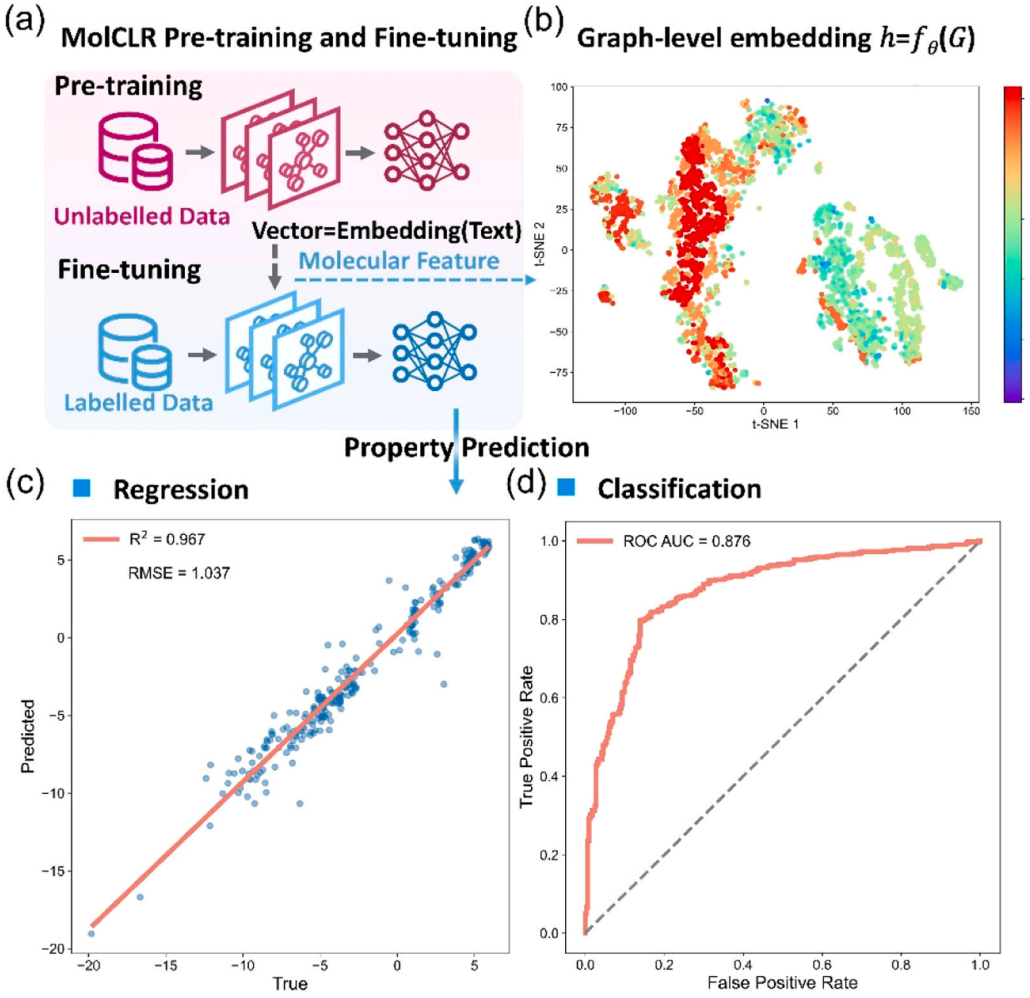


Fig. 3. Deep learning-enabled prediction of compound properties for semantic knowledge augmentation. (a) MolCLR pre-training and fine-tuning strategy for molecular feature extraction. In the pre-training stage, a graph neural network encoder is trained with a contrastive learning objective on a large set of unlabelled molecules from PubChem to learn graph-level molecular embeddings. In the fine-tuning stage, the same encoder is initialized with the pre-trained weights and optimized on labelled environmental and health risk datasets to predict eight physicochemical properties and mutagenicity for each compound. (b) Two-dimensional t-SNE visualization of the learned graph-level embeddings. Each point represents a compound, the x- and y-axes correspond to the first and second t-SNE dimensions. (c) Regression performance on downstream tasks. The solid line indicates the ideal $y = x$ relationship. (d) Classification performance on downstream tasks. The receiver operating characteristic (ROC) curve plots the true positive rate (y-axis) against the false positive rate (x-axis) over varying decision thresholds.

3.2. Predicting physicochemical properties to support reasoning

Health risk reasoning often requires detailed compound-level data, including physicochemical properties that influence exposure pathways, bioaccumulation, and toxicological outcomes. However, many compounds, especially those found in occupational or poorly regulated environments, lack experimental data. To address this limitation, we introduced a deep learning-based property prediction module to enrich our health risk knowledge base with domain-relevant attributes.

Our approach used a pretraining–fine-tuning framework based on the MolCLR model (Fig. 3a). Pretraining on large molecular datasets allowed the model to learn transferable chemical features. These features were then adapted to predict eight key properties relevant to environmental health: hydrogen bond acceptor count, topological polar surface area (TPSA), vapor pressure (PLV), aqueous solubility (SW), partition coefficients (KOC, KH), lipophilicity (Lipo), and mutagenicity (Muta). These properties were selected based on their importance in health risk profiles and their relevance to exposure scenario.

To validate the model's chemical representations, we visualized the learned embeddings using t-SNE (Fig. 3b). Structurally related compounds formed distinct clusters, suggesting that the model effectively captured meaningful molecular relationships. Partial overlaps indicated subtle differences, possibly reflecting variations in structure-activity dynamics.

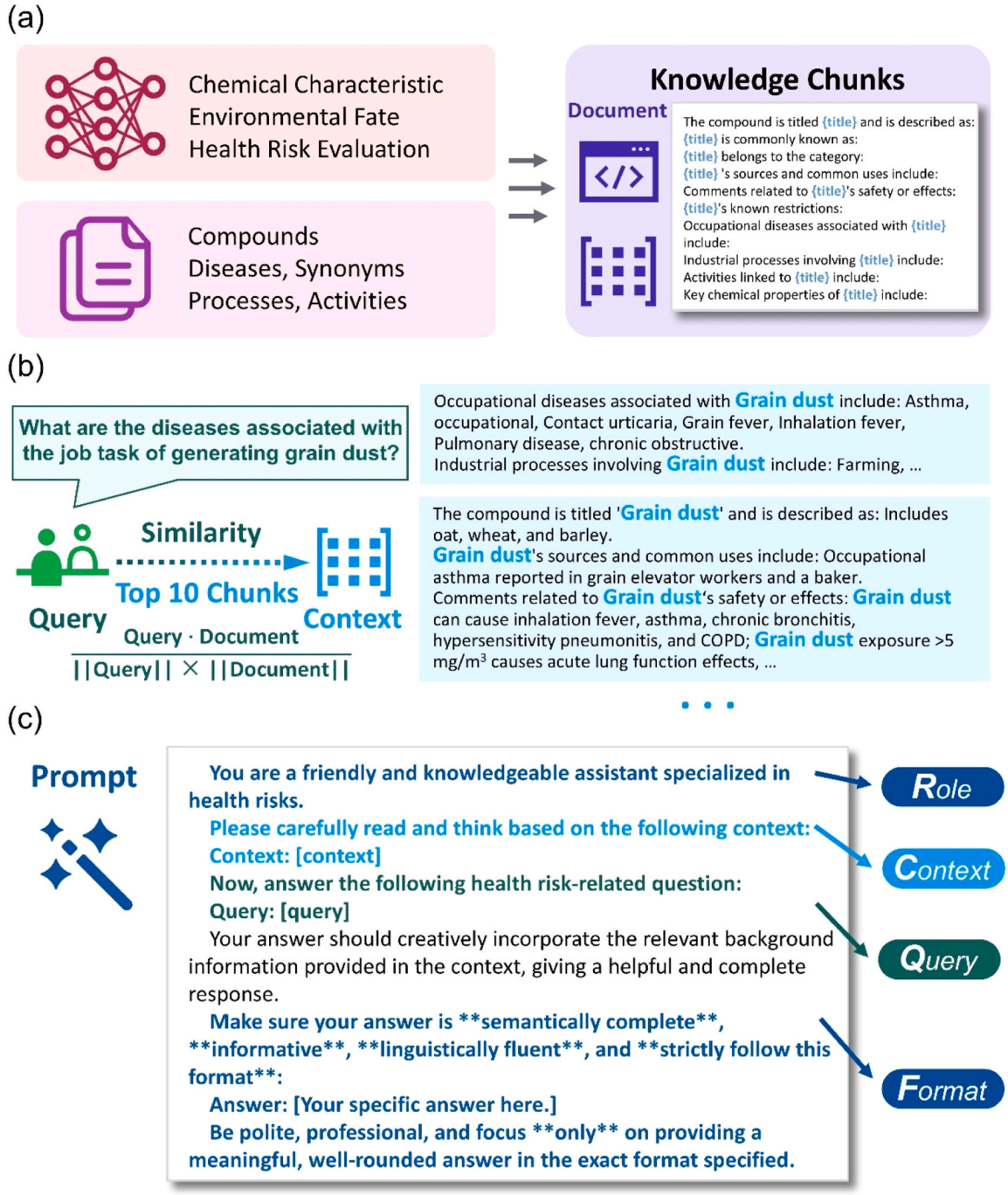


Fig. 4. Semantic structuring and prompt design for knowledge-guided reasoning. (a) Key elements of the semantically restructured knowledge base. (b) Retrieval of the top 10 context-relevant chunks based on semantic similarity to the user query. (c) Structured prompt design incorporating role, context, format, and query components to guide domain-specific reasoning in LLM-RAG.

The prediction module showed strong performance across tasks. KH prediction achieved an R^2 of 0.96, with KOC, PLV, and Lipo also exceeding 0.7 (Fig. 3c, Fig. S2a-d). RMSE values were low, confirming the robustness of the fine-tuned model and indicating that predictions closely aligned with real molecular properties in the test set. In a classification task for mutagenicity, the model reached an AUC of 0.876 (Fig. 3d), demonstrating its ability to distinguish pathogenic from non-pathogenic compounds.

These results affirm that the learned molecular descriptors effectively capture critical chemical information, enhancing property prediction and offering potential applicability across domains in physical chemistry, environmental endpoints and human toxicity. By incorporating these predictive properties into the knowledge base, we enhanced the LLM's ability to reason about chemical safety in the absence of complete empirical data. The component supports risk estimation across a wide spectrum of environmental and occupational exposure scenarios, especially for compounds with sparse toxicological characterization.

3.3. Semantic structuring for reliable retrieval in complex domain scenarios

To improve the reliability of LLM responses in complex health risk scenarios, we introduce a knowledge engineering strategy, namely semantic structuring. Semantic structuring reorganizes semi-structured domain data into task-adaptive and semantically coherent representations optimized for retrieval-augmented generation.

The original health risk knowledge base consisted of two major components: (1) numerical data on chemical properties, environmental behavior, and pathogenic potential derived from deep learning and cheminformatics; (2) unstructured or loosely structured text describing compounds, exposure activities, industrial processes, and associated diseases. Direct embedding of such mixed-format data often led to semantic misalignment, where retrieved content lacked contextual completeness or precision. Semantic structuring addresses this by reorganizing compound-level entries into self-contained and uniformly formatted knowledge chunks that better reflect LLM input–output expectations.

After semantic structuring, each chunk follows a standardized template. It includes the compound name, description, chemical category, common uses, exposure contexts, toxicological effects, known restrictions, occupational disease links, relevant industrial processes, and predicted physicochemical properties (Fig. 4a). The design ensures both interpretability and retrieval relevance. We implemented the chunking and retrieval system using LangChain's VectorStore module. Textual data was embedded into a high-dimensional vector space, and the system retrieved the top-k entries based on cosine similarity and a tuned relevance threshold. This enabled high-fidelity matching between user queries and context documents.

As an example, in response to a query on long-term exposure risks and safety controls for 2,4,5,2′,4′,5′-hexabromobiphenyl (Q17), the system retrieved documents covering bioaccumulation potential, carcinogenicity, occupational hazard profiles, and industrial exposure pathways (Fig. 4b). Minor semantic mismatches were still observed, such as references to structurally similar compounds (e.g., 3,3′,4,4′,5,5′-Hexachlorobiphenyl). However, most retrieved documents provided robust context, including exposure limits, incident analyses, and process-specific risks.

Critically, this restructuring process reduced hallucinations by improving contextual coherence and semantic fidelity. Even in cases of incomplete or noisy source data, Semantic structuring aligned generated outputs with verified knowledge representations. The results demonstrate that deliberate data organization substantially enhances retrieval accuracy and output reliability, forming a key component of the overall health risk reasoning framework.

3.4. Domain-aware prompting strategy for LLM responses

Health risk questions often require prompts that emphasize task clarity, logical flow, and standardized output. Effective prompt design is essential to ensure that LLMs generate accurate, relevant, and context-sensitive responses. We developed a domain-aware prompting strategy to reduce hallucinations, maintain task focus, and adapt to diverse exposure scenarios. The design was iteratively refined through extensive testing in health risk reasoning tasks and is illustrated in Fig. 4c. The prompt template consists of four components: role, context, question, and format. Each component serves a distinct control function.

Role prompting establishes the AI's identity as a domain expert in chemical and environmental health. The framing aligns the model's reasoning style with professional norms, reinforcing topic relevance and scientific tone. By anchoring the response generation in a domain-specific role, the model prioritizes health-related knowledge and adopts an authoritative communication style. The role prompting strategy reduces off-topic drift and improves the precision of generated outputs. Unlike generic role-playing, the approach emphasizes domain fidelity over conversational realism, ensuring responses are specialized and policy-aligned.

While role prompting defines the model's domain identity, effective reasoning also requires structured and context-aware inputs. Context prompting, built upon our semantic structuring strategy, activates task-relevant knowledge by supplying semantically coherent compound profiles drawn from the curated knowledge base. These prompts embed key molecular attributes, such as TPSA, KH, KOC, and mutagenicity, enabling the model to infer risk indicators like bioaccumulation potential, toxicity levels, and exposure thresholds across acute and chronic scenarios.

To support multi-dimensional reasoning, chain-of-thought (CoT) patterns are embedded within the prompt to guide logical progression. It allows the model to connect chemical features to exposure pathways, toxicokinetic properties, and regulatory considerations. By leveraging context enriched through semantic structuring, the prompting strategy ensures that heterogeneous data streams are synthesized into consistent, interpretable, and scientifically faithful responses.

Once the semantic context is established, question prompting guides the model to extract and reason over the most relevant information. Prompts are explicitly structured to reflect the analytical goals of health risk assessment, ensuring alignment with predefined scientific objectives. Building on the structured context, CoT frameworks embedded in the question prompt enable the model to decompose complex queries into interpretable reasoning steps. For example, in evaluating neurotoxicity and chronic hepatic injury risks from Zirconium 1-propoxide in solvent mixtures, the model is guided through a staged process: first linking chemical properties to exposure routes, then prioritizing toxicity endpoints, and finally assessing long-term risks based on bioaccumulation and mutagenicity indicators. The structure not only improves reasoning transparency but also anchors the model's attention on medically and environmentally actionable insights.

To ensure the interpretability and consistency of generated outputs, format prompting provides structural scaffolding for LLM responses. The component balances guidance with generative flexibility, particularly in open-ended or scenario-specific queries. While zero-shot prompting enables adaptability, soft formatting constraints direct the model to maintain analytical clarity and professional tone. Prompts include flexible directives such as comprehensive analysis of question elements, integration of quantitative evidence (e.g., exposure thresholds, toxicity indices), temporal framing of risks (acute versus chronic), and suggestions for mitigation strategies. These design choices promote responses that are not only informative and prescriptive, but also aligned with scientific expectations. By shaping both structure and style, format prompting minimizes hallucinations and enhances the model's ability to generalize across diverse environmental and occupational health contexts. Together, these four prompt components form a cohesive strategy

that enables the LLM to reason over complex, multi-dimensional health risk scenarios with accuracy, consistency, and domain fidelity. Although our work focuses on chronic and scenario-based chemical health risk assessment, the same prompting structure could in principle be adapted to other environmental health tasks. This would involve changing the role description and question while reusing the context and format scaffolds.

3.5. Dual evaluation reveals the value of structuring and prompting

To systematically assess model performance in health risk reasoning, we employed two complementary evaluation strategies: RAGAS and LLM-as-a-judge. RAGAS was first applied to the original 40-question deep-reasoning benchmark to quantify faithfulness and answer relevancy by analyzing the alignment between generated responses and the retrieved knowledge base context. For the subsequent judge-based assessment, we expanded the benchmark to 100 questions (40 deep-reasoning questions plus 60 broad-coverage questions) to strengthen robustness and chemical-class coverage. In parallel, the LLM-as-a-judge framework offered human-aligned scoring on correctness, utility, and informativeness, using GPT-4o-mini as an independent evaluator to minimize bias. The evaluation protocol instructed the judge model to ignore surface cues such as response length and position to ensure impartiality.

All six LLM/LLM-RAG variants achieved high faithfulness scores (0.91–0.97), but their answer relevancy varied substantially (0.78–0.90) (Fig. 5a-b). The full LLM-RAG system, which incorporates semantic structuring and structured prompting, attained the highest relevancy (~0.9), indicating its superior ability to extract and utilize context-specific knowledge in risk reasoning. In contrast, the variant without semantic structuring exhibited the lowest relevancy (<0.8), revealing that unstructured or fragmented knowledge impairs semantic coherence and task-specific interpretation. Notably, the faithfulness score remained relatively high even without semantic structuring, suggesting that hallucinations were reduced. However, the model struggled to pinpoint the most relevant information, confirming the role of well-structured context in precise reasoning.

The RAGAS metrics successfully captured surface-level alignment but were limited in discriminating performance in complex health scenarios. They lacked sensitivity to failure cases involving incomplete mechanistic pathways, missing numerical reasoning, or poor contextual synthesis. For example, RAGAS could not fully penalize responses that were faithful to context but incomplete or shallow in toxicological interpretation. By contrast, the LLM-as-a-judge framework captured subtle reasoning deficits and domain-specific incoherence, demonstrating stronger resolution and generalizability for open-ended health risk queries (Fig. 5c).

Using the LLM-as-a-judge framework, we conducted a detailed evaluation across 100 benchmark questions (Text S5-S44). The full LLM-RAG model achieved the highest mean score (27.03/30), outperforming the baseline LLM ($\Delta = 2.54$) and all ablated variants (Fig. 5d). Importantly, it also achieved the highest rank in 86 out of 100 questions (Fig. 5e), demonstrating not only superior average performance but also consistency across question types. The variant without semantic structuring, despite an average score of 24.64, showed the largest performance variance (standard deviation = 3.02), indicating unstable reasoning when dealing with unfamiliar chemical scenarios or multi-step exposure questions. This confirms that context fragmentation introduces response volatility and undermines reasoning reliability.

Prompt ablation further revealed the necessity of each design element. Removing role specification led to a consistent drop in the LLM-as-a-judge score (on the order of 2.75 points on the 30-point scale), mainly affecting professional tone and the targeting of occupational and environmental details. Omitting explicit format guidance led to a similar moderate decline of 2.82 points and yielded less structured, more numerically ambiguous responses. By contrast, simplifying the context component produces the largest effect, with a reduction of 12.4% relative to the full model, because the model loses access to the richer and semantically structured profiles that support multi-step reasoning. These patterns are consistent across the nine question categories.

In a small subset of questions (Text S5, S10), the full LLM-RAG model did not achieve the top score. These lower-ranked cases were mainly associated with three factors. These included multi-endpoint queries requiring compact synthesis of risks and controls, limited scenario-specific occupational process information in the retrieved context, and data scarcity for certain compounds. These observations suggest that expanding domain-specific evidence coverage is a direct path to further improving context specificity without changing the overall framework.

Performance by question category further validated the need for integrated strategies. The full LLM-RAG model maintained high performance across all nine thematic categories. On the contrary, baseline and ablated models underperformed in areas requiring cumulative reasoning, exposure estimation, or occupational specificity. In Question 13 (Text S17), which compares chronic toxicity between Sulfaquinoxaline and arsenic or fluoride, only models with full context and CoT prompting offered mechanistic insight into toxicokinetics and relative bioaccumulation. In Question 40 (Text S44), focused on occupational asthma from grain dust exposure, variants lacking role or contextual prompts failed to mention threshold limits or occupation-specific risks.

To further validate scientific validity beyond automated judging, we conducted a manual expert review and compared expert ratings with the AI-judge scores (Fig. S3). Human-in-the-loop validation showed strong alignment between the LLM-as-a-judge scores and the aggregated expert ratings (Pearson's $r = 0.80$, $p < 0.001$), supporting that the automated judge metric reflects expert consensus in our setting.

Together, these results indicate that hallucination mitigation and domain reasoning quality do not arise from knowledge structuring, retrieval, or prompting alone. Rather, they depend on the deliberate integration of these components. Semantic structuring anchors the model's inputs in relevant, coherent knowledge, which improves answer relevancy and response stability. Domain-aware prompting then orchestrates those inputs into logically consistent, interpretable outputs. This integrated, multi-layered knowledge reasoning framework offers a scalable blueprint for trustworthy LLM applications in environmental health, occupational exposure, and chemical safety domains. At the same time, the current implementation has been optimized and evaluated only for chemical exposures in environmental and occupational settings. Therefore, it should be regarded as a decision-support tool that complements, rather than replaces, conventional risk-assessment practice and expert judgement. Furthermore, the architecture is modular and could in principle be extended to other risk factors. However, this would require hazard-specific structured knowledge profiles and task-relevant exposure metrics.

4. Conclusions

Our integrated LLM-RAG framework combines semantic structuring and domain-aware prompting to enable accurate and context-specific reasoning in environmental health. By transforming compound data into coherent profiles and guiding outputs with task-specific prompts, the system effectively links chemical properties, exposure pathways, and disease risks. The structured approach improves answer relevancy, consistency, helpfulness, and toxicological insight, as confirmed by retrieval-based and AI-judgment-based evaluations. Beyond mitigating hallucinations, the system functions as an interpretable decision-support tool across public and occupational health domains. Its modular design supports adaptation to other applications requiring mechanistic reasoning, such as chronic exposure analysis and compound prioritization under data scarcity. With expanded data inputs and computing power, the system could evolve into a memory-enabled and self-adaptive agent capable of real-time health risk assessment. Our work demonstrates that combining semantic structuring with domain-aware

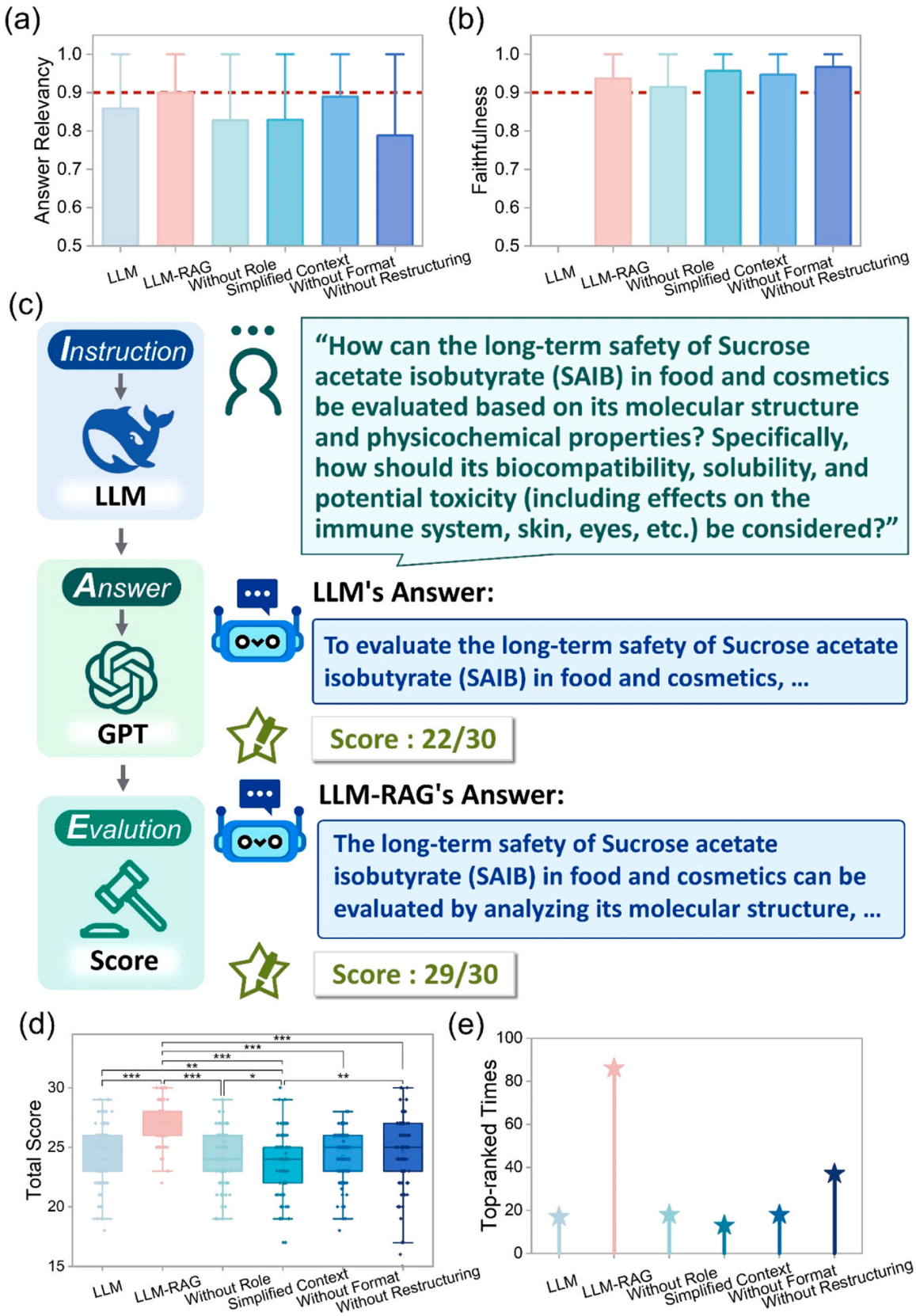


Fig. 5. Dual evaluation of LLM assistants highlights the impact of structuring and prompting. (a) Answer relevancy scores from RAGAS. (b) Faithfulness scores from RAGAS. (c) Illustration of the LLM-as-a-judge workflow, comparing LLM and LLM-RAG answers to the same question. (d) Distribution of total LLM-as-a-judge scores across the 100 benchmark questions. The x-axis lists the six assistant variants, and the y-axis shows the total score per answer (0−30); significance is annotated by brackets and asterisks (two-sided paired *t*-test; *p < 0.05, **p < 0.01, ***p < 0.001). (e) Number of questions for which each assistant achieved the top LLM-as-a-judge score, including those ranked first or within one point.

prompting offers a practical path toward building trustworthy and task-aligned LLMs for environmental health. The combination may also support reliable decision-making across complex exposure contexts.

Environmental Implication

Chemical risks, especially from rare or poorly studied hazardous chemicals, span diverse occupational and everyday scenarios, where structured knowledge is often incomplete and poorly integrated. In this work, we present a health risk assistant that integrates deep learning-based chemical property prediction, retrieval-augmented generation and prompt engineering into a unified reasoning framework. This framework establishes a scalable and trustworthy approach for AI-supported health risk assessment of hazardous chemicals in environmental and occupational contexts, and offers a pathway for developing advanced tools for public and health protection.

CRediT authorship contribution statement

Chen Qian: Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Methodology, Investigation, Conceptualization. **Han-Qing Yu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Zhuo Chen:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Meng Du:** Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank the National Natural Science Foundation of China (52170056, 52192684, 22422608 and 52293441), and the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China (JYB2025XDXM909) for supporting this work.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2026.141820.

Data Availability

Data will be made available on request.

References

[1] Wang, H., Fu, T., Du, Y., Gao, W., Huang, K., Liu, Z., et al., 2023. Scientific discovery in the age of artificial intelligence. Nature 620, 47–60. https://doi.org/10.1038/s41586-023-06221-2.
[2] Muratov, E.N., Bajorath, J., Sheridan, R.P., Tetko, I.V., Filimonov, D., Poroikov, V., et al., 2020. QSAR without borders. Chem Soc Rev 49, 3525–3564. https://doi.org/10.1039/D0CS00098A.
[3] Tropsha, A., Isayev, O., Varnek, A., Schneider, G., Cherkasov, A., 2024. Integrating QSAR modelling and deep learning in drug discovery: the emergence of deep QSAR. Nat Rev Drug Discov 23, 141–155. https://doi.org/10.1038/s41573-023-00832-0.
[4] Zang, X., Zhao, X., Tang, B., 2023. Hierarchical molecular graph self-supervised Learning for property prediction. Commun Chem 6, 34. https://doi.org/10.1038/s42004-023-00825-5.
[5] Fang, Y., Zhang, Q., Zhang, N., Chen, Z., Zhuang, X., Shao, X., et al., 2023. Knowledge graph-enhanced molecular contrastive learning with functional prompt. Nat Mach Intell 5, 542–553. https://doi.org/10.1038/s42256-023-00654-0.
[6] Fang, X., Liu, L., Lei, J., He, D., Zhang, S., Zhou, J., et al., 2022. Geometry-enhanced molecular representation learning for property prediction. Nat Mach Intell 4, 127–134. https://doi.org/10.1038/s42256-021-00438-4.
[7] Zhang, Z., Cui, P., Zhu, W., 2022. Deep learning on graphs: a survey. IEEE Trans Knowl Data Eng 34, 249–270. https://doi.org/10.1109/TKDE.2020.2981333.
[8] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Lio', Y. Bengio, Graph Attention Networks, ArXiv abs/1710.10903; 2017.
[9] A. Vaswani, N.M. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, ; 2017. Attention is All you Need.
[10] Huang, R., Xia, M., Sakamuru, S., Zhao, J., Shahane, S.A., Attene-Ramos, M., et al., 2016. Modelling the Tox21 10 K chemical profiles for in vivo toxicity prediction and mechanism characterization. Nat Commun 7, 10425. https://doi.org/10.1038/ncomms10425.
[11] Wang, H., Liu, W., Chen, J., Wang, Z., 2023. Applicability domains based on molecular graph contrastive learning enable graph attention network models to accurately predict 15 environmental end points. Environ Sci Technol 57, 16906–16917. https://doi.org/10.1021/acs.est.3c03860.
[12] von Borries, K., Holmquist, H., Kosnik, M., Beckwith, K.V., Jolliet, O., Goodman, J. M., et al., 2023. Potential for machine learning to address data gaps in human toxicity and ecotoxicity characterization. Environ Sci Technol 57, 18259–18270. https://doi.org/10.1021/acs.est.3c05300.
[13] Zhou, X., Zhou, X., Wang, C., Zhou, H., 2023. Environmental and human health impacts of volatile organic compounds: a perspective review. Chemosphere 313, 137489. https://doi.org/10.1016/j.chemosphere.2022.137489.
[14] Li, J., Guan, Z., Wang, J., Cheung, C.Y., Zheng, Y., Lim, L.L., et al., 2024. Integrated image-based deep learning and language models for primary diabetes care. Nat Med 30, 2886–2896. https://doi.org/10.1038/s41591-024-03139-8.
[15] Liu, T., Duan, Y., 2024. Beware the self-fulfilling prophecy: enhancing clinical decision-making with AI. Crit Care 28, 276. https://doi.org/10.1186/s13054-024-05062-3.
[16] Singhal, K., Azizi, S., Tu, T., Mahdavi, S.S., Wei, J., Chung, H.W., et al., 2023. Large language models encode clinical knowledge. Nature 620, 172–180. https://doi.org/10.1038/s41586-023-06291-2.
[17] Wan, P., Huang, Z., Tang, W., Nie, Y., Pei, D., Deng, S., et al., 2024. Outpatient reception via collaboration between nurses and a large language model: a randomized controlled trial. Nat Med 30, 2878–2885. https://doi.org/10.1038/s41591-024-03148-7.
[18] Habicht, J., Viswanathan, S., Carrington, B., Hauser, T.U., Harper, R., Rollwage, M., 2024. Closing the accessibility gap to mental health treatment with a personalized self-referral chatbot. Nat Med 30, 595–602. https://doi.org/10.1038/s41591-023-02766-x.
[19] Boiko, D.A., MacKnight, R., Kline, B., Gomes, G., 2023. Autonomous chemical research with large language models. Nature 624, 570–578. https://doi.org/10.1038/s41586-023-06792-0.
[20] M. Bran, A., Cox, S., Schilter, O., Baldassari, C., White, A.D., Schwaller, P., 2024. Augmenting large language models with chemistry tools. Nat Mach Intell 6, 525–535. https://doi.org/10.1038/s42256-024-00832-8.
[21] Jablonka, K.M., Schwaller, P., Ortega-Guerrero, A., Smit, B., 2024. Leveraging large language models for predictive chemistry. Nat Mach Intell 6, 161–169. https://doi.org/10.1038/s42256-023-00788-1.
[22] Kang, Y., Kim, J., 2024. ChatMOF: an artificial intelligence system for predicting and generating metal-organic frameworks using large language models. Nat Commun 15, 4705. https://doi.org/10.1038/s41467-024-48998-4.
[23] Tian Gong, AI for Sustainability. ⟨https://www.tiangong.earth/ai⟩; 2023 (accessed 28 August 2024).
[24] Zhu, J.-J., Yang, M., Jiang, J., Bai, Y., Chen, D., Ren, Z.J., 2024. Enabling GPTs for expert-level environmental engineering question answering. Environ Sci Technol Lett 11, 1327–1333. https://doi.org/10.1021/acs.estlett.4c00665.
[25] Wu, Y., Xu, M., Liu, S., 2024. Generative artificial intelligence: a new engine for advancing environmental science and engineering. Environ Sci Technol 58, 17524–17528. https://doi.org/10.1021/acs.est.4c07216.
[26] Farquhar, S., Kossen, J., Kuhn, L., Gal, Y., 2024. Detecting hallucinations in large language models using semantic entropy. Nature 630, 625–630. https://doi.org/10.1038/s41586-024-07421-0.
[27] Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., et al., 2023. Survey of hallucination in natural language generation. Article 248. ACM Comput Surv 55. https://doi.org/10.1145/3571730.
[28] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, , A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions, ArXiv abs/2311.05232; 2023.
[29] Wang, Y., Wang, J., Cao, Z., Barati Farimani, A., 2022. Molecular contrastive learning of representations via graph neural networks. Nat Mach Intell 4, 279–287. https://doi.org/10.1038/s42256-022-00447-x.
[30] Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., et al., 2018. PubChem 2019 update: improved access to chemical data. Nucleic Acids Res 47, D1102–D1109. https://doi.org/10.1093/nar/gky1033.
[31] L. Zheng, W.L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, , Judging LLM-as-a-judge with MT-Bench and Chatbot Arena, ArXiv abs/2306.05685; 2023.